

NSF NRT Project Challenges

UMKC 2026 Spring Research-A-Thon

About the NSF NRT Program

The NSF Research Traineeship (NRT) Program at UMKC supports interdisciplinary training in AI and secure networked sensing to address societal challenges such as alcohol and drug abuse. The program integrates AI, data science, and sensing technologies for real-world impact.

NSF NRT Team

- PI: Dr. Mostafizur Rahman
- Co-PIs: Dr. Yugyung Lee, Dr. Dianxiang Xu

Mission

Develop AI-driven and sensing-based systems that transform raw data into interpretable, actionable insights with ethical and responsible use.

Project Challenges

- Challenge 1 (AI): Substance Abuse Risk Detection from Social Signals
- Challenge 2 (Engineering): Vapor-Node IoT Monitoring System

Hands-on experience in AI modeling, data intelligence, embedded systems, and real-time sensing.

Research-A-Thon Event

- Website: [\[Research-A-Thon Website Link\]](#)
- Date: April 10, 2026
- Time: 9:00 AM – 12:00 PM
- Location: Student Union (SU) Room 401

Key Dates & Deliverables

Team Registration

- Deadline: April 1, 2026
- Form: [\[Registration Link\]](#)

Submission (April 6, 2026, 12:00 PM)

- Report (max 4 pages)
- Video (max 3 minutes)
- GitHub repository (well-documented code)
- Submission: [\[Submission URL\]](#)

Finalist Announcement

- April 8, 2026 (via email)

Final Live Demo

- April 10, 2026 | 9:00–10:00 AM | SU 401
- In-person system demonstration

Awards

- Up to \$2,000 total prizes
- Participation & award certificates

Learning Outcomes

- LLMs, embeddings, and RAG
- End-to-end AI pipelines
- Temporal and signal analysis
- Explainable AI systems
- Practical applications & decision support
- Ethical AI and communication

Contact

- AI Challenge: Dr. Yugyung Lee — leeyu@umkc.edu
- Engineering Challenge: Dr. Mostafizur Rahman — rahmanmo@umsystem.edu

Challenge 1: AI for Substance Abuse Risk Detection from Social Signals

Lead: Dr. Yugyung Lee

The Mission

Substance abuse is a growing societal crisis, and early warning signals often emerge in digital environments such as social media and online communities.

In this challenge, you will design an AI-driven system that identifies risk signals related to alcohol and drug abuse from public, anonymized data. The goal is to transform unstructured data into interpretable, actionable insights that support early awareness and intervention.

Datasets (Provided or Recommended)

- CDC overdose and public health data: <https://data.cdc.gov/>
- NIDA data: <https://nida.nih.gov/research-topics/trends-statistics>
- Kaggle dataset: <https://www.kaggle.com/datasets/jessicali9530/kuc-hackathon-winter-2018>
- Optional additional datasets (with approval)

Core Tasks

1. Risk Signal Detection

Identify:

- Substance use mentions
- Emotional distress or relapse signals

Expanded Expectations:

- Develop a robust pipeline to extract relevant textual signals from noisy, real-world data
- Design prompts or fine-tuning strategies for LLMs that capture nuanced language (e.g., indirect references, slang, emotional tone)
- Compare multiple approaches (e.g., rule-based vs embedding-based vs LLM-based) and analyze their performance

Using:

- Large language model prompting or fine-tuning
- Embedding-based classification or retrieval

2. Temporal and Behavioral Analysis

Detect:

- Spikes and trends
- Evolving narratives

Expanded Expectations:

- Model temporal dynamics of risk signals (e.g., sudden spikes, gradual increases, recurring patterns)
- Cluster users/posts/topics to uncover behavioral patterns and emerging themes
- Analyze how discussions evolve over time and identify potential early-warning indicators

Using:

- Embedding similarity or clustering
- Time-aware analysis

3. Explainability and Reasoning

Provide:

- Supporting evidence
- Summaries
- Interpretable outputs

Expanded Expectations:

- Generate explanations that justify model predictions using retrieved evidence or attention patterns
- Provide concise summaries for decision-makers (e.g., public health analysts)
- Ensure outputs are transparent, auditable, and aligned with ethical AI principles

Innovation Tracks (Select One)

Track A: AI Modeling and Reasoning

Focus on building a strong AI pipeline:

- Large language model-based classification
- Retrieval-augmented generation
- Embedding-based pipelines
- Explainable outputs

Track B: Data Intelligence and Decision Support

Focus on insights and system development:

- Embedding-based trend discovery
- Behavioral pattern analysis
- Interactive dashboards (Streamlit/web apps)
- Visualization of risk signals over time

Related Projects and Inspiration

Public Health Data

- <https://data.cdc.gov>
- https://www.cdc.gov/nchs/state-stats/deaths/drug-overdose.html?utm_source=chatgpt.com
- https://www.cdc.gov/nchs/nvss/vsrr/drug-overdose-data.htm?utm_source=chatgpt.com
- <https://www.cdc.gov/drugoverdose/>
- <https://monitoringthefuture.org>

Trend Analysis

- <https://trends.google.com>

Research Example

- <https://arxiv.org/abs/2004.07654>

Evaluation Criteria (Challenge 1)

- Technical quality: 40%
- Innovation: 30%
- Impact and relevance: 20%
- Clarity and communication: 10%

Important Constraints (Challenge 1)

- Use only public/provided datasets
- Do not identify individuals
- Emphasize privacy and ethics
- Focus on population-level insights

Challenge 2: Vapor-Node IoT Monitoring System

Lead: Dr. Mostafizur Rahman

The Mission

As environments become more complex, the ability to monitor invisible chemical changes is vital.

This challenge requires designing a **Distributed IoT Sensing System** that detects “volatile spikes” using low-cost sensors and real-time communication.

The Technical Core

Sensor Node (Edge)

- Arduino Uno
- MQ-3 / MQ-135 gas sensor

Expanded Expectations:

- Calibrate the sensor to distinguish meaningful signal variations from environmental noise
- Experiment with sensor placement and airflow conditions to improve detection sensitivity
- Capture continuous readings and preprocess signals at the edge

Gateway (Hub)

- Raspberry Pi

Expanded Expectations:

- Implement reliable communication between edge and hub (Serial, I2C, or MQTT)
- Process streaming sensor data in real time
- Store and visualize readings for monitoring and analysis

Innovation Tracks (Select One)

Track A: Insight Specialist (Data & Edge AI)

- Signature recognition (event spike vs baseline drift)
- Signal stabilization (Kalman filter or moving average)

Expanded Expectations:

- Design algorithms that robustly detect sudden “event spikes”
- Reduce noise caused by environmental fluctuations
- Evaluate detection accuracy under different conditions

Track B: UI/UX Specialist (Interface & Notification)

- Remote alerts (Telegram, Discord, email triggers)
- Web dashboard (Node-RED, Streamlit, Flask)

Expanded Expectations:

- Build real-time alerting systems with configurable thresholds
- Design intuitive dashboards that show historical trends and system health
- Ensure usability and clarity for non-technical users

Starter Kits Provided

- [MQ gas sensor](#) (Qty: 1)
- [Adafruit ADC](#) (Qty: 1)

Deliverables & Timeline (Challenge 2)

Final Demo

- Live “Vapor Event” simulation demonstrating detection

Report (Max 4 pages) must include:

- Circuit schematics
- Sensing strategy
- Source code (well-commented)

Judging Criteria (Challenge 2)

- System reliability (40%)
- Innovation quality (40%)

- Documentation & presentation (20%)

Final Note

These challenges align with NSF NRT goals by integrating **AI, secure sensing, and interdisciplinary research** to address critical societal challenges.